BCS2540: Numerical Methods. (Martijn Boussé)

WHY num. methods?

* approximate. ~ impossible.
* faster.
* simpler
* larger data.

1) example. from calculus: $\int x \cdot e^{-x^2} dx \xrightarrow{u=-x^2} \frac{1}{2}\int e^u du.$

$\int e^{-x^2} dx$?

↳ Impossible,

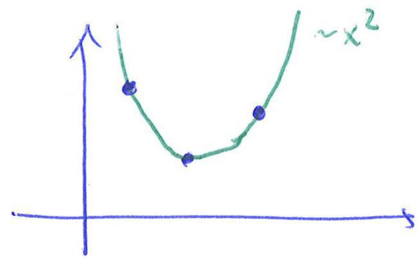
or

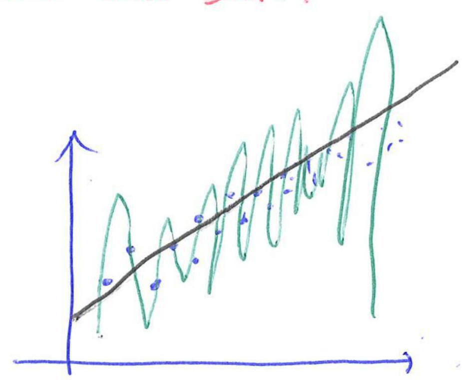
↳ integrand only known in samples.

↳ easier.

⇒ Topic 4: Num. integration and diff.

we have samples.





⇒ Topic 3: poly. interpolation.

→ data (Taylor, Newton, Lagrange)

→ function (interpol. nodes, (splines)).

⇒ Topic 5: Approximation.

→ data LS, DFT.

→ function. Fourier, ...

2) example from calculus:

$$\text{Solve } y' = k \cdot y. \xrightarrow{\text{int.}} y = A \cdot e^{kt}$$

$$y' = \sin(xy)$$

⇒ Topic 2: ODE solvers.

example from LA:

Find eigenvalues of $\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$.

$\det(A - \lambda I) = 0$ char. polynomial!

$$\det \begin{pmatrix} 1-\lambda & 2 \\ 1 & 2-\lambda \end{pmatrix} = 0$$

$$\lambda^2 - 3\lambda = 0.$$

$$\lambda(\lambda - 3) = 0$$

$\lambda = 0$ $\quad$ $\lambda = 3$

$\Rightarrow$ Topic 6: NLA.

$\Rightarrow$ Topic 1: Computer Arithmetic + Algebraic Eqs.

Before we can do anything else, how do I compute on a computer.

How do I represent numbers?

Course specifics.

6 topics → 6 modules on Canvas

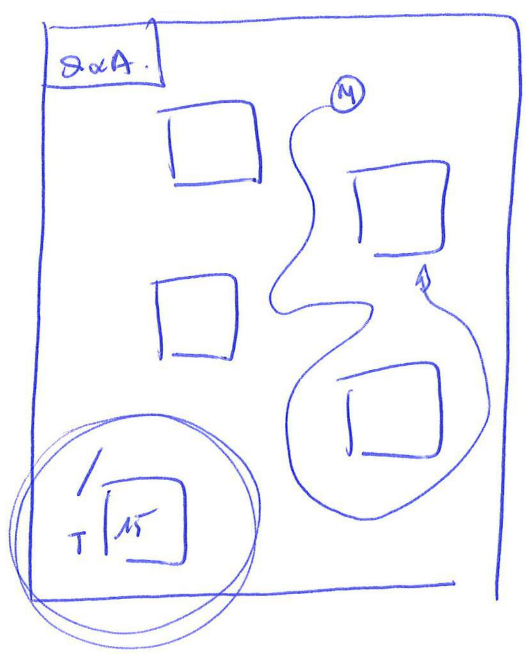
PLT

L → Frontal teaching

PTT] x2

T

GQ. → C1.005 + C1.015.



A1 Exam. 70%

A2 Matlab practical. 15%

A3 Quizzes. 15%

$\boxed{\text{passing norm of 5.5 !}}$

(week 5 / 1st of May / 11h00)

A4 Bonus 10%

↳ (1) 8 out of 10 tutorial — prep. PTT / present. / Active.

↳ (2) onboarding quiz.

↳ (3) Matlab onramp.

] week 2

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if PN AND BR

if PN AND $\overline{BR}$

if $\overline{PN}$, FG = EG.

T1A: Computer Arithmetic

⇒ How can we represent numbers on a computer? (digitally).

↑ up   1

↓ down.   0

42 in 8-bit binary: $(0 0 1 0 1 0 1 0)_2$.

But we need $\mathbb{R}$ numbers?

$\boxed{\text{Floating point representation}}$ → see handbook.

single prec (IEEE): 32 bits in FP

double prec ("): 64-bits in FP. ← Matlab.

$$\overbrace{\pm}^{1\text{bit}} \underbrace{1, XXXX\cdots X}_{1+52\text{ bit mantissa}} \cdot 2^{\overbrace{\pm XX\cdots X}^{1+10\text{ bits}}}$$

example: 1+7 bit mantissa, 1+3 bit exponent.

$$3 = +1{,}5 \cdot 2^{+1} = +1,(0000\,101)_2 \cdot 2^{+(001)_2}$$

$$1/3 = 0{,}3333\ldots = +1,(\underbrace{0101010}\,|\,1\,\cancel{0\,1}\cdots)_2 \cdot 2^{-(010)_2}$$

! by using a FP repr, we already intro an error.

Where do they come from?

→ ROUNDOFF error: due to the use of inexact arithmetic for computations (i.e., eg., FP)

→ TRUNCATION errors: due to the use of inexact method

TY: $f'(x) = \cancel{\lim_{h\to 0}} \dfrac{f(x+h) + f(x)}{h}$

$\approx$

↳ "cost": truncation error $O(h^2)$.

(DATA errors: due to measurement noise/error.)

How do we measure them?

p = exact
$\tilde{p}$ = approx.

Abs. error $= |p - \tilde{p}|$ ⟶ estimate: $|p_n - p_{n-1}|$

Rel. error $= \dfrac{|p - \tilde{p}|}{|p|}$.

Specify?

Accuracy: Accurate to n digits if n digits are correct.

Precision: is number of digits used.

$\pi = 3,1415926 \ldots$

$\tilde{p} = \underbrace{\overbrace{3,14}^{\text{3 digit accurate}}\,2857\,1}_{\text{8 digits precision}},$

dp
sf.

Working guidelines
- Use a higher precision for intermed. calculations
- Use the precision appropriate for the accuracy.
- Use at most 2 sf when reporting errors.

ROUNDED ARITHMETIC. → very typical exam question.

↳ Why? To show the effect of rounding (on paper/hand calculations)
↳ How? Choose digit + round after every step

~ effect of FP representation

$$\pi \cdot e = 3{,}14159\ldots \cdot 2{,}71828\ldots =$$
$$= 8{,}53973\ldots$$
$$= 8{,}54 \ \boxed{(3sf)}$$ need to mention this
$$= 8{,}540 \quad (4sf)$$
$$= 8{,}54 \quad (2dp)$$

3-digit rounded A:

$$\pi \cdot e = \underbrace{3{,}14}_{3sf} \cdot \underbrace{2{,}72}_{3sf} = \underbrace{8{,}5408}_{exact} = 8{,}54 \ (3sf)$$

2-digit RA:

$$\pi \cdot e = \underbrace{3{,}1}_{2sf} \cdot \underbrace{2{,}7}_{2sf} = 8{,}37 = 8{,}4 \ (2sf)$$

---

given: $f(x) = x^3 - 5{,}34x^2 + 1{,}52x + 4{,}61$.
Evaluate f at $x = 4{,}89$ using 3-digit RA

ROUND AFTER EVERY operation

Atomic op. vs seq powers.

$$x^2 = x \cdot x = 4{,}89 \cdot 4{,}89 = \underbrace{23{,}9121}_{NOT} = 23{,}9 \ (3sf)$$
$$x^3 = x^2 \cdot x = 23{,}9 \cdot 4{,}89 = 116{,}871 = 117 \ (3sf)$$
$$5{,}34 \cdot x^2 = 5{,}34 \cdot 4{,}89^2 = 5{,}34 \cdot 23{,}9 = 127{,}626 = 128 \ (3sf)$$
$$1{,}52 \cdot x = \quad -- \quad = 7{,}43 \ (3sf)$$

$$f(x) = ((x^3 - 5{,}34x^2) + 1{,}52x) + 4{,}61$$
$$= ((117 - 128) + 7{,}43) + 4{,}61$$
$$= (-11 + 7{,}43) + 4{,}61$$
$$= -3{,}57 + 4{,}61$$
$$= \boxed{1{,}04} \quad \Leftrightarrow \quad f(4{,}89) = 1{,}28 \ (3sf)$$ double prec

$$149{,}905 - 146{,}936 = 2{,}969 \overset{6sf}{=} 2{,}96900$$ → $\boxed{\text{Check.}}$

→ tutorial: quadratic eq → ABC formula

$$x = \frac{\sqrt{b^2-4ac} - b}{2a}$$

or

$$x = \frac{-2c}{\sqrt{b^2-4ac} + b}$$

→ polyn. eval in <u>nested form</u>

$$x^3 - 5{,}34\,x^2 + 1{,}52\,x + 4{,}61$$

$$= ((x-5{,}34)\cdot x + 1{,}52)\,x + 4{,}61$$

Algebraic Eqs

Example: $\boxed{\text{Say I want to compute } \sqrt{a}}$ for given a.

How?

$$x = \sqrt{a}$$

$$x^2 = a$$

$$\boxed{\underbrace{x^2 - a}_{f(x)} = 0}$$

$\boxed{\text{trying to solve } f(x) = 0}$

general: given a continuous func $f: \mathbb{R} \to \mathbb{R}$ and $a, b \in \mathbb{R}$ and $a < b$, solve $f(x) = 0$ for $x \in [a, b]$.

Solution? A value p s.t. $f(p) = 0$

"root"

Bisection $\longrightarrow$ here!

Secant

Newton-Raphson

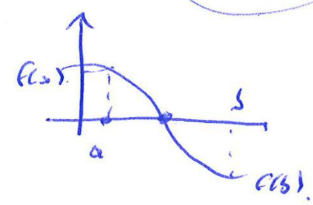
book,

Bisection

Problem: Find a root of f given bracket $[a,b]$

$\nearrow$ C!

Idea: Shrink the bracket.
while $\boxed{\text{preserving the root.}}$

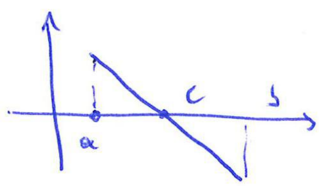


$\downarrow$

if $\text{sign}(f(a)) \neq \text{sign}(f(b))$
i.e. $f(a) \cdot f(b) < 0$.

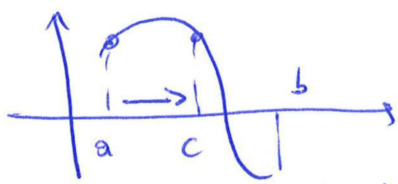
Method: Compute $c = \frac{a+b}{2}$ (midpt) of $[a,b]$.

↳ if $f(c) = 0$, then c is a root.



↳ signs must differ.

* if $\text{sign}(f(a)) = \text{sign}(f(c)) \neq \text{sign}(f(b))$, then $a \leftarrow c$



* if $\text{sign}(f(a)) \neq \text{sign}(f(c)) = \text{sign}(f(b))$, then $b \leftarrow c$.

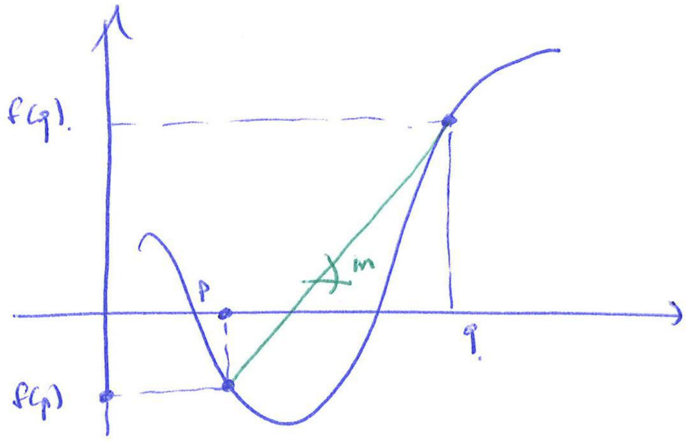
! the width of the interval is <u>halved</u>.

Secant Method

Same problem: Find a root of f given a bracket $[p, q]$.

continuous

Idea: Approximate f by the secant line between $(p, f(p))$ and $(q, f(q))$.



derivation: slope: $m = \dfrac{f(q) - f(p)}{q - p}$

secant line: $y = f(q) + m(x - q)$

set $y = 0$ and solve for $x = r$ to find intersection with x-axis.

$$0 = f(q) + m(x - q)$$

$$x = q - \frac{1}{m} f(q)$$

$$\boxed{r = q - \frac{1}{m} f(q)}$$

iterative algorithm: start with p_0, p_1: $p_{n+1} = p_n - \dfrac{p_n - p_{n-1}}{f(p_n) - f(p_{n-1})} \cdot f(p_n)$

Watchout: p_n, p_{n+1} do not need to bracket a root!

we might loose the bracket property

When do I stop? (stopping criteria)

I want to stop when $|p_n - p| < \varepsilon$. aka converged when~

BUT: we don't know p (obviously~)

If convergence is rapid, we expect $|p_n - p| \ll |p_{n-1} - p|$

If $|p_n - p| = \frac{1}{\gamma} |p_{n-1} - p|$. with $\frac{1}{\gamma} \leq \frac{1}{2}$,

this is how I quantify the decrease.

find $|p_{n-1} - p_n| \geq |p - p_n|$.

this bounds that and this we can compute.

so 'practically' stop when $|p_n - p_{n-1}| < \varepsilon$.

! Different stopping criteria exist: fixed # iter, residual, error, ...

error estimate: for the heuristic. $|p_n - p_{n-1}| < \varepsilon$, expect $|p_n - p| \lesssim \varepsilon$.

error bound: If also $f(p_n)$ and $f(p_{n-1})$ have different signs (i.e. bracket prop is preserved)

then $|p_n - p| < \varepsilon$.

prove via IVT.

Iterative method (Matlab)

```
function r = secant(f,p,q,e).
    while abs(q-p) > e,
        r = ...;
        p = q; q = r;
    end.
end.
```

Secant: Example

Estimate $\sqrt{2}$ to accuracy of $\overset{\varepsilon}{\overset{\shortparallel}{0{,}01}}$. Start with $p_0 = 1$ and $p_1 = 2$.

Define $f(x) = x^2 - 2$ ... (same as before).

$p_0 = 1$; $f(p_0) = -1{,}000$ (3dp)
$p_1 = 2$; $f(p_1) = 2{,}000$ (3dp).

$n=1$

$$p_2 = p_1 - \frac{p_1 - p_0}{f(p_1) - f(p_0)} \cdot f(p_1) = 2{,}000 - \frac{2{,}000 - 1{,}000}{2{,}000 - (-1{,}000)} \cdot 2{,}000 = 1{,}333 \text{ (3dp)}$$

$|p_2 - p_1| = |1{,}333 - 2{,}000| = 0{,}667 > \varepsilon$
$f(p_2) = -0{,}222$. (only when you continue!)

$n=2$

$$p_3 = p_2 - \frac{p_2 - p_1}{f(p_2) - f(p_1)} \cdot f(p_2) = 1{,}333 - \frac{1{,}333 - 2{,}000}{-0{,}222 - 2{,}000} \cdot (-0{,}222) = 1{,}400 \text{ (3dp)}.$$

$|p_3 - p_2| = |1{,}400 - 1{,}333| = 0{,}067 > \varepsilon$.
$f(p_3) = -0{,}040$

$n=3$

$p_4 = \ldots = 1{,}415$
$|p_4 - p_3| = |1{,}415 - 1{,}400| = 0{,}015 > \varepsilon$.
$f(p_4) =$

$n=4$

$p_5 = \ldots = 1{,}414$
$|p_5 - p_4| = |1{,}414 - 1{,}415| = 0{,}001 < \varepsilon$. $\longrightarrow$ Stop.

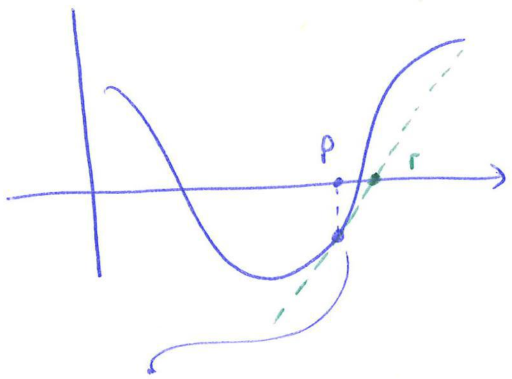
We can expect $|p_5 - \sqrt{2}| < \varepsilon$.

actual error $|p_5 - \sqrt{2}| = 2{,}1 \cdot 10^{-4} \ll 0{,}01$. $\longrightarrow$ it's a rough bound!

Newton-Rhapson: Method

Same problem: Find a root of f given a bracket $[p, q]$.

Idea: Instead of using a secant line joining $(p, f(p))$ and $(q, f(q))$, use the tangent line at $(p, f(p))$.



deriv ~ instantaneous change rather than
secant ~ avg change.

Derivation: tangent line at $(p, f(p))$ is given by $y = f(p) + f'(p)(x-p)$

set $y=0$ and solve for $x=r$ to find intercept with x-axis:

$$r = p - \frac{f(p)}{f'(p)}.$$

Iterative Alg: Start with p_0: $p_{n+1} = p_n - \frac{f(p_n)}{f'(p_n)}.$

$$\sim \quad p_{n+1} = p_n - \frac{1}{d} f(p_n).$$

vs

$$p_{n+1} = p_n - \frac{1}{m} f(p_n).$$

When do I stop? Same as for secant $|p_n - p_{n-1}| < \varepsilon$.

Matlab: (ToDo)

Newton-Raphson: Example

Estimate $\sqrt{2}$ to accuracy of $0{,}01 = \varepsilon$. Start with $p_0 = 1{,}0000$ (4dp)

Use $f(x) = x^2 - 2$. and $f'(x) = 2x$.

init. $p_0 = 1{,}0000$ (4dp)
$f(p_0) = 1{,}0000^2 - 2 = -1{,}0000$ (4dp)
$f'(p_0) = 2 \cdot 1{,}0000 = 2{,}0000$ (4dp)

$n=1$ $p_1 = p_0 - \dfrac{f(p_0)}{f'(p_0)} = 1{,}0000 - \dfrac{-1{,}0000}{2{,}0000} = 1{,}5000$ (4dp)

$|p_1 - p_0| = 0{,}50$ (2sf) $> \varepsilon$ so continue.

$n=2$ $f(p_1) = 0{,}2500$ (4dp)
$f'(p_1) = 3{,}0000$ (4dp)
$p_2 = p_1 - \dfrac{f(p_1)}{f'(p_1)} = \text{---} = 1{,}4167$ (4dp)

$|p_2 - p_1| = 0{,}083$ (2sf) $> \varepsilon$ so continue.

$n=3$ $f(p_2) = 0{,}0069$ (4dp)
$f'(p_2) = 2{,}8333$ (4dp)
$p_3 = p_2 - \dfrac{f(p_2)}{f'(p_2)} = \text{---} = 1{,}4142$ (4dp).

$|p_3 - p_2| = 0{,}0025$ (2sf) $< \varepsilon$. So we stop.

Newton-Raphson : Convergence analysis (7.1.7.6)

Let p^* be a root.

some $\xi \in (p^*, p_n)$.

1) Expand around p_n : 1) $f(p) = f(p_n) + \cdots$ 2) $f(p^*) = 0$

$$f(p^*) = f(p_n) + f'(p_n)(p^* - p_n) + \frac{1}{2} f''(\xi)(p^* - p_n)^2.$$

2) We know that $f(p^*) = 0$ because p^* is a root. ξ

$$0 = f(p_n) + f'(p_n)(p^* - p_n) + \frac{1}{2} f''(\xi)(p^* - p_n)^2.$$

$$-f(p_n) = f'(p_n)(p^* - p_n) + \frac{1}{2} f''(\xi)(p^* - p_n)^2$$

3) NR method:

$$p_{n+1} = p_n - \frac{f(p_n)}{f'(p_n)}.$$

$$p_{n+1} = \cancel{p_n} + (p^* - \cancel{p_n}) + \frac{1}{2} \frac{f''(\xi)}{f'(p_n)} (p^* - p_n)^2$$

$$\underbrace{p_{n+1} - p^*}_{= \varepsilon_{n+1}} = \frac{1}{2} \frac{f''(\xi)}{f'(p_n)} \underbrace{(p^* - p_n)^2}_{\varepsilon_n.}$$

4)

$$\varepsilon_{n+1} = \frac{1}{2} \frac{f''(\xi)}{f'(p_n)} \varepsilon_n^2$$

?

$$\varepsilon_{n+1} \leq \frac{1}{2} \frac{f''(p^*)}{f'(p^*)} \varepsilon_n^2.$$

$$\varepsilon_{n+1} = C \varepsilon_n^{2}\ \text{P?}$$

C → for some constant.

error is quadratic!
very fast
(when close to solution)

Algebraic Equations : Comparison of Methods

	Bisection	Secant	NR.
convergence guaranteed?	Yes	No	No.
bracket property	Yes	No	No.
conv speed.	linear. $\varepsilon_{n+1} \sim \varepsilon_n$	superlinear. $\varepsilon_{n+1} \sim \varepsilon_{n+1}^{1.6}$	superlin / quadratic. $\varepsilon_{n+1} \sim \varepsilon_n^2$
requirements.	+/−	f.	f and f'!

!

Regula Falsi

Brent's / Dekker method.

Idea: compute next step using bisection AND secant. then use the step that improves the interval the most. but take care to keep the bracket property as you go.

no bracket prop → hence use ~~[illegible]~~ Po & P2 instead of P1 & P2